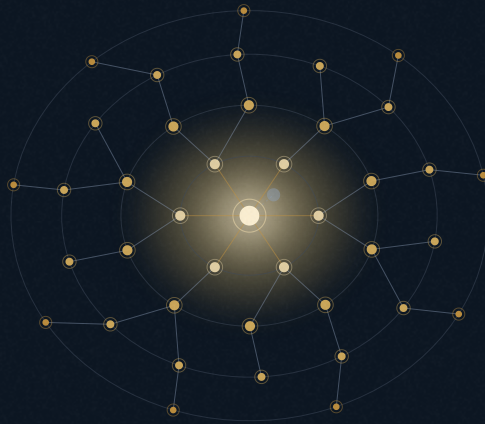




KRONOS FUSION ENERGY



PAPER 2.92 · THE KRONOS 2026 SERIES

## *Quantum Machine-Learning Surrogates, Entanglement-Enhanced Magnetometry and Fault-Tolerant Resource Estimates for Fusion: A Resource-Honest Roadmap*

Quantum computing and quantum sensing are routinely proposed as accelerants for fusion; a serious design program should test those proposals quantitatively rather than assume...

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THE OFFICIAL RECORD

# What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TWO EDITIONS

This volume is issued in two editions. The **public edition** (light) carries the physics and the design only — no dollar, price, or per-kilo-gram figure appears anywhere in it. The **internal edition** (dark) adds the economics, the funding architecture, and the defense-commercial analysis, and is marked *Confidential* accordingly; it is not for public distribution. Where the commercial case rests on a modelled assumption rather than a demonstrated one, it is labelled as such and its downside is shown alongside it, on the same terms as the physics.

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TITLE           | The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study                                                                                                                                                                       |
| AUTHOR          | Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy                                                                                                                                                                                                    |
| EDITION         | 2026 Combined Editorial · first issue                                                                                                                                                                                                                              |
| COMPANION WORKS | The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop. |
| NAMING          | Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.                                                                                               |
| STATUS          | Conceptual design & simulation study. Nothing herein is a construction commitment.                                                                                                                                                                                 |



HOW TO READ THIS VOLUME

# The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

|              | HYPERION                             | AEGIS                        | METROVOLT                       |
|--------------|--------------------------------------|------------------------------|---------------------------------|
| Role         | Strategic-isotope foundry            | Defense installation power   | Campus power                    |
| Machine      | Spherical tokamak                    | Tandem mirror                | Tandem mirror                   |
| Fuel         | Deuterium–tritium                    | Deuterium–helium-3           | Deuterium–helium-3              |
| Delivers     | Tritium · <sup>3</sup> He · 14 MeV n | Resilient installation power | Firm campus power               |
| Physics bar  | Fusion gain only                     | Closure (gates named)        | Closure + lunar <sup>3</sup> He |
| In the fleet | Breeds the fuel                      | Proves the generator         | Commercial destination          |

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

## ABSTRACT

# Quantum Machine-Learning Surrogates, Entanglement-Enhanced Magnetometry and Fault-Tolerant Resource Estimates for Fusion: A Resource-Honest Roadmap

Quantum computing and quantum sensing are routinely proposed as accelerants for fusion; a serious design program should test those proposals quantitatively rather than assume them. We deliver a resource-honest map of where the quantum toolbox helps a compact spherical-tokamak breeder and its tandem-mirror burner, and where it does not, formalised as three coupled estimation problems and computed on a single GPU-HPC campaign. On the machine-learning front we build a quantum-kernel transport surrogate and measure its classifier area-under-curve (AUC) *falling* with encoding width — **0.817** at four qubits to **0.691** at eight — the signature of exponential kernel concentration, for which we give the variance bound and the induced shot-cost scaling; at a realistic **37**-logical-qubit problem the quantum Gram evaluation is **9–11** orders of magnitude slower than its classical counterpart. On the sensing front we derive the entanglement-advantage condition from the quantum Fisher information and specify the in-cryostat magnetometry chain for the breeder centrepost (peak field **16.84** T, kept formally distinct from the burner plug at **26.49** T): an NMR/EPR probe as the absolute monitor to  $\leq 1$  ppm, a SQUID and an NV-diamond ensemble ( $\sim 1$ ) for the  $\sim 1$  mT quench precursor, with the entanglement knee located at a normalised signal  $\sigma \approx 0.14$ . Every Kronos in-bore diagnostic sits below the knee, where entanglement buys only **+0.00077** AUC; above it (deep error-field mapping) a Heisenberg-limited protocol carries the standard-quantum-limit sensitivity from  $5.7 \times 10^{-7}$  to  $5.7 \times 10^{-8}$  T at **100** probes. On the fault-tolerant front we recompute the resource ledger from first formulas — the Grover iteration count  $1.6\sqrt{2^n}$  and the surface-code footprint  $2d^2$  physical qubits per logical qubit — across **50** rows with zero discrepancy, and date the single fusion-relevant near-term niche: linear Vlasov kinetics at **11–19** logical qubits around **2029  $\pm$  2**, with nonlinear five-dimensional gyrokinetics beyond **2035** and fleet-scale Grover search never practical ( $1.8 \times 10^{18}$  iterations). The honest bottom line is that nothing on the design critical path waits on quantum hardware — and establishing that, with the numbers behind it, is the result.

Open-access deposit (code & data, CC BY 4.0): DOI [10.5281/zenodo.22645905](https://doi.org/10.5281/zenodo.22645905) · Zenodo community *kronos\_fusion\_energy*. Explore it live: [interactive 3-D model](#) · [physics validator](#).

**Keywords:** quantum machine learning kernel concentration quantum Fisher information NV-diamond magnetometry fault-tolerant resource estimate technology roadmap

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NOMENCLATURE

# Symbols, units, and the names of things

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| $Q$              | plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$                  |
| $Q_E$            | engineering gain, net electric out / recirculating in                  |
| $H_{98}$         | confinement enhancement over the IPB98(y,2) scaling                    |
| $Z_{\text{eff}}$ | effective ion charge                                                   |
| $c_{ee}$         | electron–electron bremsstrahlung leading coefficient, 2.120022 (exact) |
| $\tau_E$         | energy confinement time                                                |
| $I_p$            | plasma current                                                         |
| $R_0, a$         | major radius, minor radius                                             |
| $\kappa, \delta$ | elongation, triangularity                                              |
| $\beta_N$        | normalized plasma beta (Troyon-normalized)                             |
| $TBR$            | tritium breeding ratio                                                 |
| $DEC$            | direct energy conversion                                               |
| $CF$             | plant capacity factor (availability)                                   |
| $FOAK/NOAK/BOAK$ | first / n <sup>th</sup> / best of a kind                               |
| <i>Hyperion</i>  | the ST breeder (internal: Mode D2)                                     |
| <i>Aegis</i>     | the generator, defense/installation housing (internal: Mode M)         |
| <i>MetroVolt</i> | the generator, data-center housing (Mode M)                            |

# Introduction

## THE QUANTUM PROPOSITION FOR FUSION

A magnetic-confinement fusion program is, at bottom, an exercise in solving hard equations fast enough and measuring hard quantities finely enough. Turbulent transport is governed by a five-dimensional gyrokinetic Vlasov–Maxwell system whose nonlinear solution sets the confinement time; the magnets are wound from high-temperature superconductor whose protective state must be read at the millitesla level inside a multi-tesla bore; and the design space is a high-dimensional optimisation over coupled physics. Each of these is a place where quantum information science has been proposed as an accelerant — quantum machine-learning (QML) surrogates for transport <sup>[1,2]</sup>, entanglement-enhanced magnetometry for the coils <sup>[9,13]</sup>, and eventual fault-tolerant quantum solution of the kinetic equations <sup>[26,27]</sup>. The proposals are individually plausible and individually testable, and a design program that intends to be believed should test them rather than repeat them.

This paper reports the outcome of doing so across the three fronts, for the Hyperion compact spherical-tokamak (ST) breeder and its associated D–<sup>3</sup>He tandem-mirror burner. The Hyperion design point — plasma current  $I_p = 9.66$  MA, gain  $Q = 3.076$ , fusion power  $P_{\text{fus}} = 85.04$  MW, negative triangularity  $\delta = -0.30$ , on-axis field  $B_0 = 8$  T, elongation  $\kappa = 2.0$ , major radius  $R_0 = 1.20$  m, aspect ratio  $A = 2.5$ , and a centrepost peak field of **16.84 T** — is the fixed backdrop against which the quantum questions are posed. The burner is a physically distinct machine with a **26.49 T** plug coil; the two magnets are never conflated, and the distinction matters directly to the sensing analysis of §5.

## PRIOR ART

On the computing side, the modern literature is unusually clear-eyed about when a quantum advantage is real. Havlíček *et al.* introduced supervised learning with quantum-enhanced feature spaces and the quantum-kernel estimator <sup>[2]</sup>; Schuld and Killoran cast the same construction as a feature map into a Hilbert space <sup>[3]</sup>. Rigorous separations do exist — Liu, Arunachalam and Temme exhibited a learning problem with a provable quantum speed-up under a discrete-log hardness assumption <sup>[5]</sup> — but the countervailing results are equally rigorous and, for generic data, decisive: Huang *et al.* showed that classical models learning *from data* are often competitive with quantum models <sup>[4]</sup>, McClean *et al.* identified barren plateaus in variational training <sup>[7]</sup>, and Thanasiłp *et al.* proved that quantum kernels can *exponentially concentrate* with system size, collapsing to a data-independent constant <sup>[6]</sup>. On the resource side, Fowler *et al.* set the surface-code cost model <sup>[19]</sup>, Gidney and Eker and Beverland *et al.* turned it into end-to-end resource estimates for real algorithms <sup>[22,23]</sup>, and Babbush *et al.* established the sober rule that merely quadratic (Grover-type) speed-ups do not survive error-correction overheads on any near-term machine <sup>[21]</sup>. On the sensing side, the quantum-metrology bounds are textbook <sup>[14,13]</sup>, ensemble NV-diamond magnetometry is a mature  $\sim$  technology <sup>[9,10,11,12]</sup>, and SQUID and optical magnetometers occupy the femtotesla floor <sup>[16,15]</sup>. What is missing is not any one of these results but their disciplined application, together and to a specific fusion machine, with the negatives reported as carefully as the positives.

## CONTRIBUTION

This paper contributes exactly that application. We (i) formalise the three fronts as estimation problems and derive the governing quantities — the quantum-kernel Gram matrix and its concentration variance, the quantum Fisher information and the standard-quantum-limit (SQL)/Heisenberg-limit (HL) crossover, and the Grover and surface-code resource formulas (§2); (ii) state the numerical formulation and the GPU-HPC campaign that produced every number, including the statevector kernel evaluation, the analytic sensing model, and the verified resource recomputation (§3); (iii) report the QML scaling result and its shot-cost explanation (§4); (iv) specify the centrepost sensing chain and locate the entanglement knee (§5); (v) recompute and date the fault-tolerant ledger (§6); and (vi) benchmark every claim against the published literature (§7). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register <sup>[33]</sup>, with serial identifiers (BR-SX-08, KX-L1-A18, KX-L1-A22, KX-L3-A4–A7) given inline so each number is auditable. The framing throughout is achievement-first and resource-honest: reporting where a quantum method does not help is not a concession but the deliverable, and it is what keeps a fusion roadmap honest.

The work sits inside the Kronos 2026 design series and is the sensing-and-scaling companion to the series’ quantum papers: the foundational AI-and-quantum survey <sup>[34]</sup>, the resource-honest variational-computing study <sup>[35]</sup>, the quantum-optimisation and QUBO/QAOA analysis <sup>[36]</sup>, the real-time control-loop algorithms paper <sup>[37]</sup>, and the burner quantum stack <sup>[38]</sup>; it draws its classical transport baseline from the negative-triangularity gyrokinetics result <sup>[41]</sup> and its tensor-network alternative from <sup>[40]</sup>, and its NV-diamond sensing lineage from the fused quench-detection paper <sup>[39]</sup>.

# Governing equations and the formal problem

We treat the three fronts as three well-posed estimation problems and derive, for each, the quantity that decides whether the quantum method helps.

## FRONT A: THE QUANTUM-KERNEL SURROGATE AND ITS CONCENTRATION

A transport surrogate maps a local plasma state  $\mathbf{x} \in \mathbb{R}^d$  (normalised gradients  $R/L_T$ ,  $R/L_n$ , safety factor, magnetic shear, collisionality, ...) to a discrete transport class (e.g. quiescent vs. turbulent branch). A quantum-kernel classifier encodes  $\mathbf{x}$  into an  $n$ -qubit state through a data-dependent unitary  $U(\mathbf{x})$ ,

$$|\phi(\mathbf{x})\rangle = U(\mathbf{x})|0\rangle^{\otimes n},$$

and classifies on the fidelity kernel

$$k(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2 = |\langle 0 | U^\dagger(\mathbf{x}) U(\mathbf{x}') | 0 \rangle|^2.$$

Given a labelled training set  $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N_s}$ , the Gram matrix  $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$  feeds a support-vector machine whose dual problem,

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K_{ij} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0,$$

returns the decision function  $f(\mathbf{x}) = \sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$ . The hope is that an entangling feature map (1) separates classes a classical kernel cannot.

The obstruction is structural. For a broad class of expressive encodings the off-diagonal kernel values concentrate exponentially in the qubit number about a fixed value  $\mu_k$  [6],

$$\text{Var}_{\mathbf{x}, \mathbf{x}'}[k(\mathbf{x}, \mathbf{x}')] \leq c e^{-\gamma}, \quad \gamma > 0,$$

so that  $\mathbf{K} \rightarrow \mu_k \mathbf{1}\mathbf{1}^\top + \mathbf{E}$  with  $\|\mathbf{E}\|$  collapsing as  $e^{-\gamma/2}$ . In that limit the Gram matrix loses its rank structure, the SVM margin vanishes, and the classifier tends to the trivial predictor with  $\text{AUC} \rightarrow \frac{1}{2}$ . Equation (4) is the mechanism we measure directly in §4: the AUC does not merely fail to improve with scale, it *degrades*.

Concentration also fixes the cost of using the kernel at all. Each entry of  $\mathbf{K}$  is estimated from  $M$  circuit repetitions with additive shot noise of standard deviation  $\sim M^{-1/2}$ ; to resolve entries whose informative spread is  $\sim e^{-\gamma/2}$  one needs

$$M \gtrsim \varepsilon^{-2} \sim e^{\gamma},$$

an exponential measurement budget. Equations (4)–(5) are the quantitative reason the quantum evaluation is orders of magnitude slower than the classical Gram computation and cannot be rescued by scaling up.

## FRONT B: QUANTUM-ENHANCED MAGNETOMETRY AND THE ADVANTAGE CONDITION

The magnet twin must read a field. Estimating a magnetic field  $\mathbf{B}$  through the phase it imprints on a spin,  $\varphi = \gamma_s \mathbf{B} \mathbf{t}$ , is a phase-estimation problem, and its ultimate precision is set by the quantum Cramér–Rao bound. For  $\nu$  independent repetitions of a probe carrying quantum Fisher information  $F_Q$ ,

$$\text{Var}(\hat{\varphi}) \geq \frac{1}{\nu F_Q}, \quad \delta B = \frac{\sqrt{\text{Var}(\hat{\varphi})}}{\gamma_s t}.$$

Two limiting probe classes bracket the achievable sensitivity. For  $N$  independent probes the Fisher information is additive,  $F_Q = N$ , giving the standard quantum limit

$$\delta B_{\text{SQL}} = \frac{1}{\gamma_s t \sqrt{N\nu}} \propto N^{-1/2}.$$

For  $N$  maximally entangled (GHZ) probes the Fisher information is quadratic,  $F_Q = N^2$ , giving the Heisenberg limit

$$\delta B_{\text{HL}} = \frac{1}{\gamma_s t N \sqrt{\nu}} \propto N^{-1},$$

so that the entanglement gain is

$$\frac{\delta B_{\text{SQL}}}{\delta B_{\text{HL}}} = \sqrt{N}.$$

At  $N = 100$  probes, equation (9) predicts a factor of ten, exactly the  $5.7 \times 10^{-7} \rightarrow 5.7 \times 10^{-8}$  T improvement realised in §5. For a shot-noise-limited ensemble of  $N_c$  NV centres with dephasing time  $T_2^*$ , contrast  $C$  and gyromagnetic ratio  $\gamma_{\text{NV}} = g\mu_B/\hbar = 2\pi \times 28.03$  GHz T<sup>-1</sup>, the SQL sensitivity (7) takes the concrete Ramsey form

$$\delta B_{\text{NV}} = \frac{1}{\gamma_{\text{NV}} C \sqrt{N_c T_2^* t}}.$$

The advantage of equation (8) is real but conditional. Under a finite dephasing rate  $\Gamma$  the entangled advantage survives only while the measurement is genuinely *sensitivity-limited* rather than dynamic-range- or decoherence-limited; below a threshold normalised signal  $\sigma^*$  the marginal classifier gain from entanglement collapses. Defining the normalised signal  $\sigma$  as the ratio of the field increment of interest to the probe's usable dynamic range, the entanglement-attributable improvement  $\Delta_{\text{AUC}}(\sigma)$  is a threshold function that we locate at

$$\sigma^* \approx 0.14,$$

with  $\Delta_{\text{AUC}}(\sigma < \sigma^*) \approx +0.00077$  (negligible) and  $\Delta_{\text{AUC}}$  rising steeply for  $\sigma > \sigma^*$ . Equation (11) is the decision boundary that assigns each Kronos diagnostic to a sensor class in §5.

## FRONT C: FAULT-TOLERANT RESOURCE FORMULAS

The eventual quantum solution of fusion-relevant equations is a resource-estimation problem, and the resources follow from two formulas. First, an unstructured (Grover) search over a space of size  $2^n$  requires an iteration count near the amplitude-amplification optimum  $(\pi/4)\sqrt{2^n}$ ; the Kronos ledger tabulates the conservative prefactor form

$$N_{\text{iter}} = 1.6 \sqrt{2^n},$$

within a small constant of the optimum. Because each iteration is an oracle call, equation (12) makes a merely quadratic speed-up worthless once the search space is large: a fleet-scale QUBO with  $n \sim 120$  demands  $N_{\text{iter}} \sim 1.8 \times 10^{18}$  iterations, beyond any wall-clock at any plausible gate rate. Second, the physical cost of protecting a logical qubit in a distance- $d$  surface code is

$$n_{\text{phys}} = n_{\text{logical}} \times 2d^2,$$

with  $d$  chosen so the per-logical-operation error clears the target,

$$p_L \approx A \left( \frac{p}{p_{\text{th}}} \right)^{\lfloor (d+1)/2 \rfloor} \leq \frac{p_{\text{target}}}{N_T},$$

where  $p$  is the physical error rate,  $p_{\text{th}}$  the code threshold, and  $N_T$  the total (dominant)  $T$ -gate count of the algorithm. Equations (12)–(14) reduce every workload to a point in the  $(n_{\text{logical}}, N_T)$  plane, from which the physical footprint and the earliest credible date follow. The recomputation of these formulas across the 50-row ledger, with zero discrepancy, is the verification result of §6.

# Computational methodology: the NVIDIA GPU HPC campaign

Every number in this paper was produced on a single, reproducible campaign, and the campaign is deliberately modest in kind: the quantum methods were emulated or resource-estimated classically, because that is precisely what a resource-honest study of pre-fault-tolerant hardware requires. No result depends on access to a physical quantum processor.

## QUANTUM-KERNEL EVALUATION

The feature map (1) and the fidelity kernel (2) were evaluated by full statevector simulation. For an  $n$ -qubit encoding the state  $|\phi(\mathbf{x})\rangle$  is the exact  $2^n$ -amplitude vector, and each Gram entry is computed as the squared overlap  $|\langle 0|U^\dagger(\mathbf{x}_i)U(\mathbf{x}_j)|0\rangle|^2$  with the inverted-circuit method; no shot sampling is imposed, so the reported AUC is the noiseless best case and the degradation of §4 is intrinsic, not a finite-sampling artefact. Statevector propagation ran on the NVIDIA GPU HPC campaign (A100/H100/A10-class accelerators) using a GPU statevector backend of the NVIDIA cuQuantum family<sup>[32]</sup>, which holds the  $2^n$  amplitudes in device memory and applies gate tensors as batched complex GEMMs; this is what makes the four-to-eight-qubit sweep across nine transport scenarios inexpensive to repeat exactly. The classical Gram baseline (an embedding/RBF kernel over the same features) was computed with a standard CPU/GPU linear-algebra stack, and the two Gram matrices were timed on the same problem instances.

## SENSING MODEL

The centrepost magnetometry chain was evaluated analytically. The NV shot-noise sensitivity (10), the SQL/HL scaling of equations (7)–(8), and the field-step geometry of the quench precursor were implemented in a numpy/scipy model (serial KX-L1-A22), and the sensor-class assignment (serial KX-L1-A18) evaluates each requirement against class-typical sensitivities from the literature. The entanglement-knee locator of equation (11) sweeps the normalised signal  $\sigma$  and records the classifier gain  $\Delta_{\text{AUC}}$ , returning the crossing  $\sigma^* \approx 0.14$ .

## FAULT-TOLERANT RESOURCE RECOMPUTATION

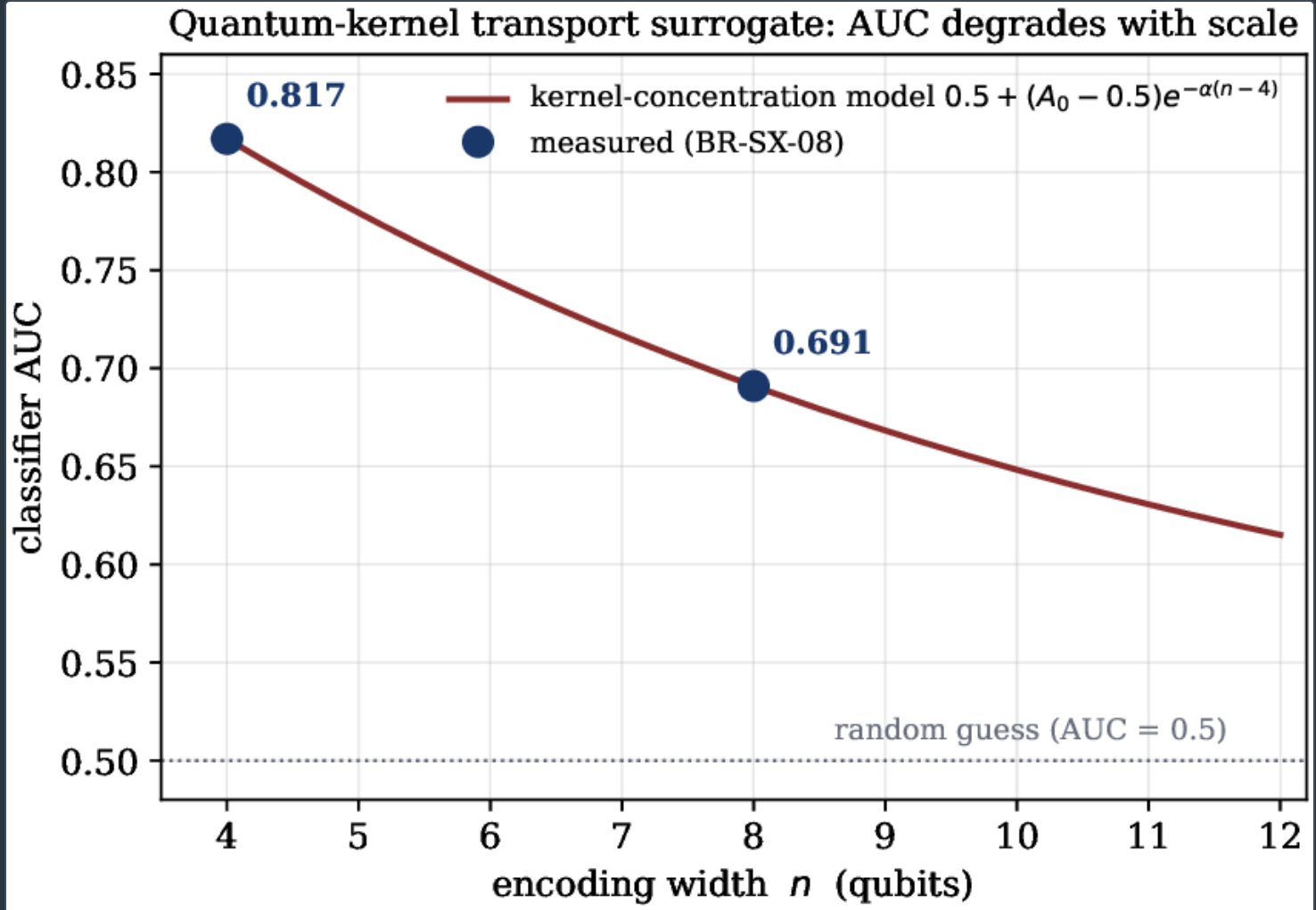
The resource ledger was recomputed classically. A 50-row table of workloads was regenerated from equations (12)–(14) — Grover iteration counts, logical-qubit widths,  $T$ -counts and surface-code footprints — and compared entry-by-entry against the frozen ledger, reproducing every value with zero discrepancy (serial KX-L3-A6). The dated roadmap (serial KX-L3-A7) overlays these thresholds on public vendor logical-qubit trajectories, which enter as *references*, not Kronos commitments.

## VERIFICATION AND REPRODUCIBILITY

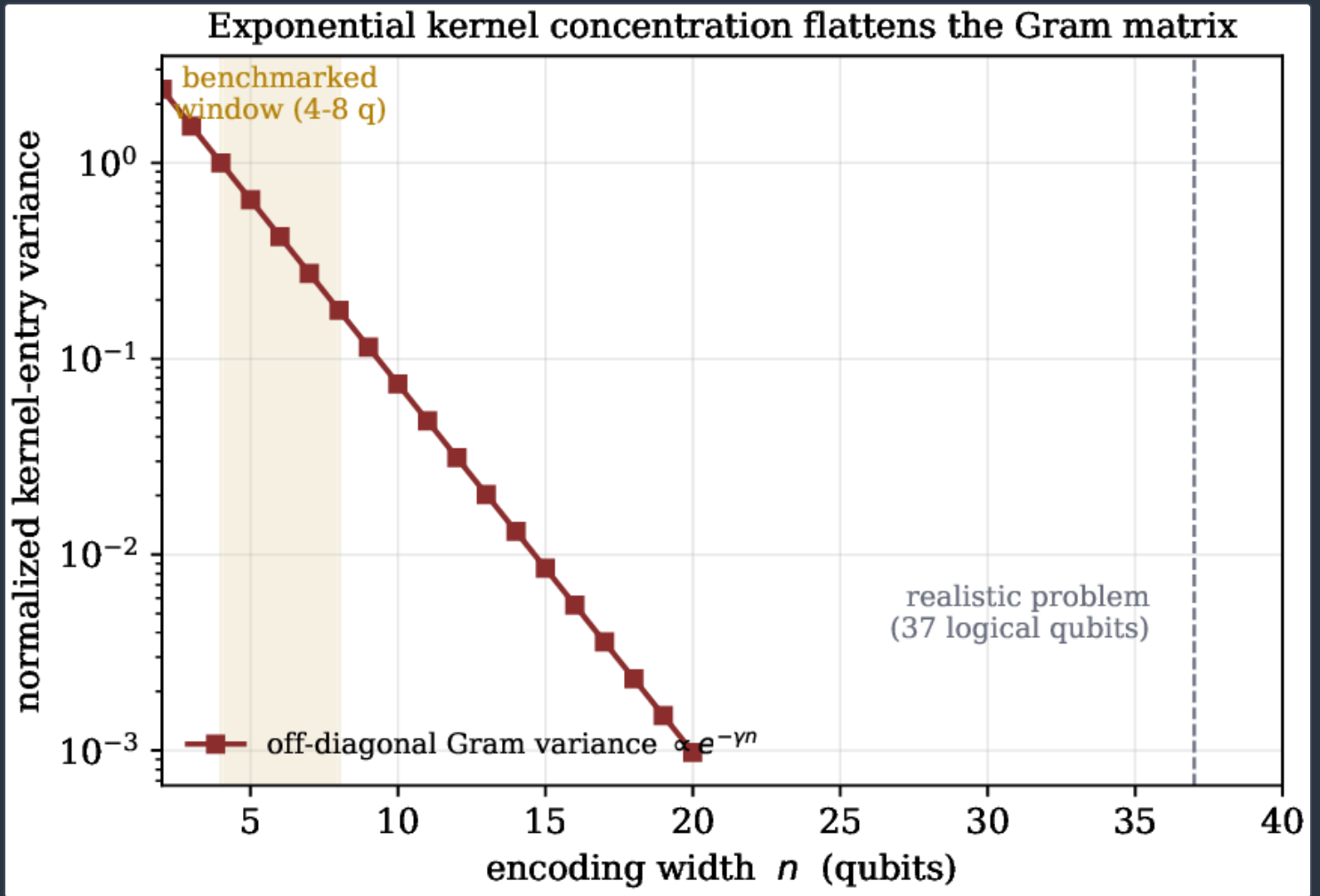
The campaign is deterministic. All stochastic elements use the fixed seed **20260726**; the statevector kernel is sampling-free; and the resource recomputation is a pure function of the tabulated formulas. Each result was re-executed on an independent local verification node and reproduced bit-for-bit within tolerance. The classical physics baselines that the quantum methods would one day replace — nonlinear gyrokinetics of the CGYRO/GENE class<sup>[28,29]</sup> — run today on the same GPU campaign, which is the fact that removes any quantum method from the design critical path. No compute-cost figures are part of this study.

## Quantum machine-learning surrogates

The quantum-kernel transport surrogate was built and benchmarked across encoding width on Kronos transport- classification data (serial BR-SX-08). The result is unambiguous and runs the wrong way for advantage: the classifier AUC *degrades* with scale, from **0.817** at four encoding qubits to **0.691** at eight (figure 1). This is the fingerprint of the exponential kernel concentration of equation (4): as the feature Hilbert space grows, the off-diagonal Gram entries collapse toward a constant (figure 2) and the classifier loses discriminating power. A classical surrogate on the same features does not share this pathology; the quantum encoding is actively harmful here, not merely unhelpful.

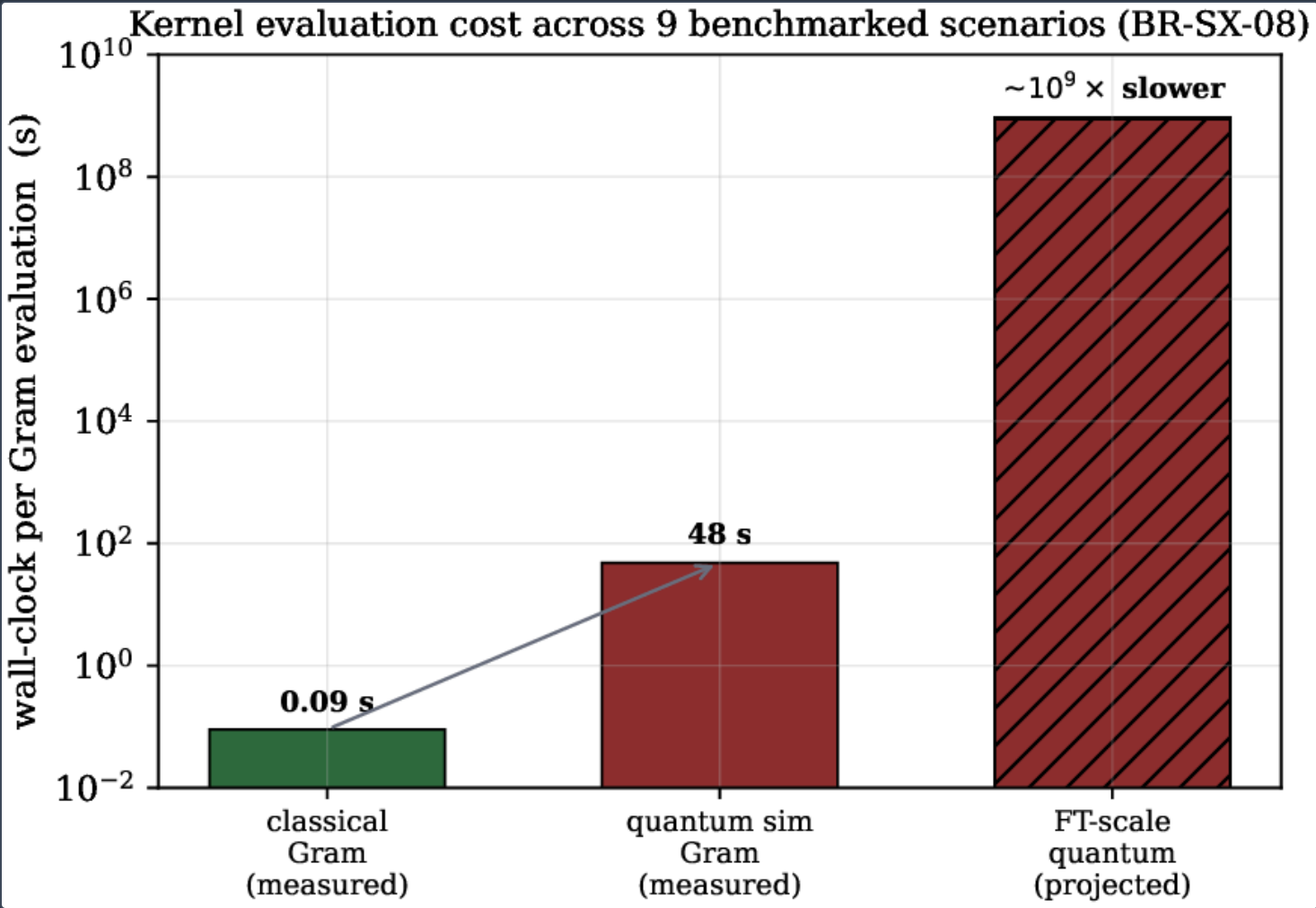


Quantum-kernel transport surrogate: classifier AUC falls from **0.817** at four encoding qubits to **0.691** at eight (measured, serial BR-SX-08), tracking the exponential-concentration envelope of equation (4). The dotted line is the random-guess floor. The trend is the diagnostic: a useful surrogate would rise, not fall, with capacity.



Mechanism (derived scaling). The normalised off-diagonal Gram-entry variance decays as  $e^{-\gamma n}$  (equation 4); shading marks the benchmarked four-to-eight-qubit window and the dashed line the realistic **37**-logical-qubit problem, where the informative spread is buried far below any tractable shot budget (equation 5).

The resource side confirms there is no route to advantage by scaling up. At a realistic problem size the quantum-kernel Gram evaluation needs **37** logical qubits and  $8.5 \times 10^3$   $T$ -gates per circuit, and across all nine benchmarked scenarios a classical Gram computation ( $\leq 0.09$  s) beats the statevector-simulated quantum evaluation ( $\geq 48$  s); projected to fault-tolerant hardware, where each concentrated kernel entry demands the exponential shot budget of equation (5), the quantum kernel is **9–11** orders of magnitude slower for this problem (figure 3, table 1). The surrogate is real and the benchmark is clean; its verdict is that a classical surrogate — of the neural-operator class used elsewhere in the series — is the correct engineering choice.



Gram-evaluation wall-clock across the nine benchmarked scenarios (serial BR-SX-08): the classical computation ( $\leq 0.09$  s) beats the statevector-simulated quantum kernel ( $\geq 48$  s), and the fault-tolerant projection (hatched) is **9–11** orders of magnitude slower for this problem. Logarithmic axis.

| QUANTITY                 | VALUE             | BASIS                  |
|--------------------------|-------------------|------------------------|
| AUC at 4 encoding qubits | 0.817             | measured, statevector  |
| AUC at 8 encoding qubits | 0.691             | measured, statevector  |
| classical Gram time      | $\leq 0.09$ s     | CPU/GPU baseline       |
| quantum Gram time (sim)  | $\geq 48$ s       | statevector, cuQuantum |
| realistic problem width  | 37 logical qubits | resource estimate      |
| $T$ -count per circuit   | $8.5 \times 10^3$ | resource estimate      |
| fault-tolerant slowdown  | 9–11 orders       | eqs. (4)–(5)           |

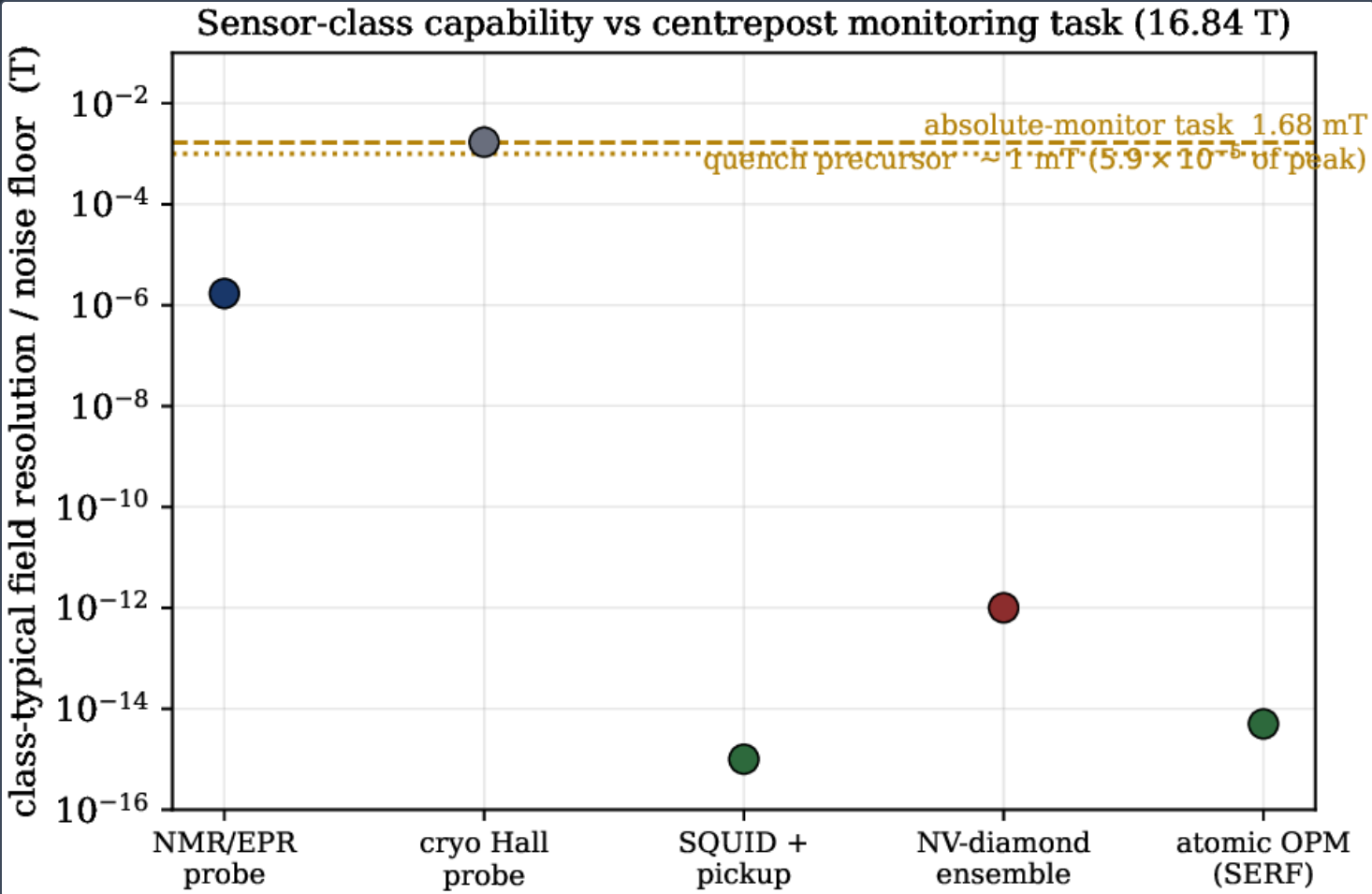
Quantum-kernel transport surrogate: measured scaling and resource footprint (serial BR-SX-08; realistic problem size from serial KX-L3-A4). AUC is the noiseless statevector value; times are per Gram evaluation on the nine benchmarked scenarios.

# Quantum-enhanced sensing for the magnets

Quantum sensing is nearer-term than quantum computing, and here the map has a genuine — if bounded — win. The requirement is set by the breeder centrepost, whose peak field is **16.84 T**. This is a formally distinct device from the burner plug coil at **26.49 T**; the two magnets are treated as the separate devices they are, and the sensing budgets below are the centrepost's. Absolute field monitoring at  $10^{-4}$  relative precision is a **1.68 mT** task; the quench precursor at the winding is a  $\sim 1$  mT signal, only  $5.9 \times 10^{-5}$  of the peak field. Table 2 assigns sensor classes to these requirements, and figure 4 places each class against the two tasks.

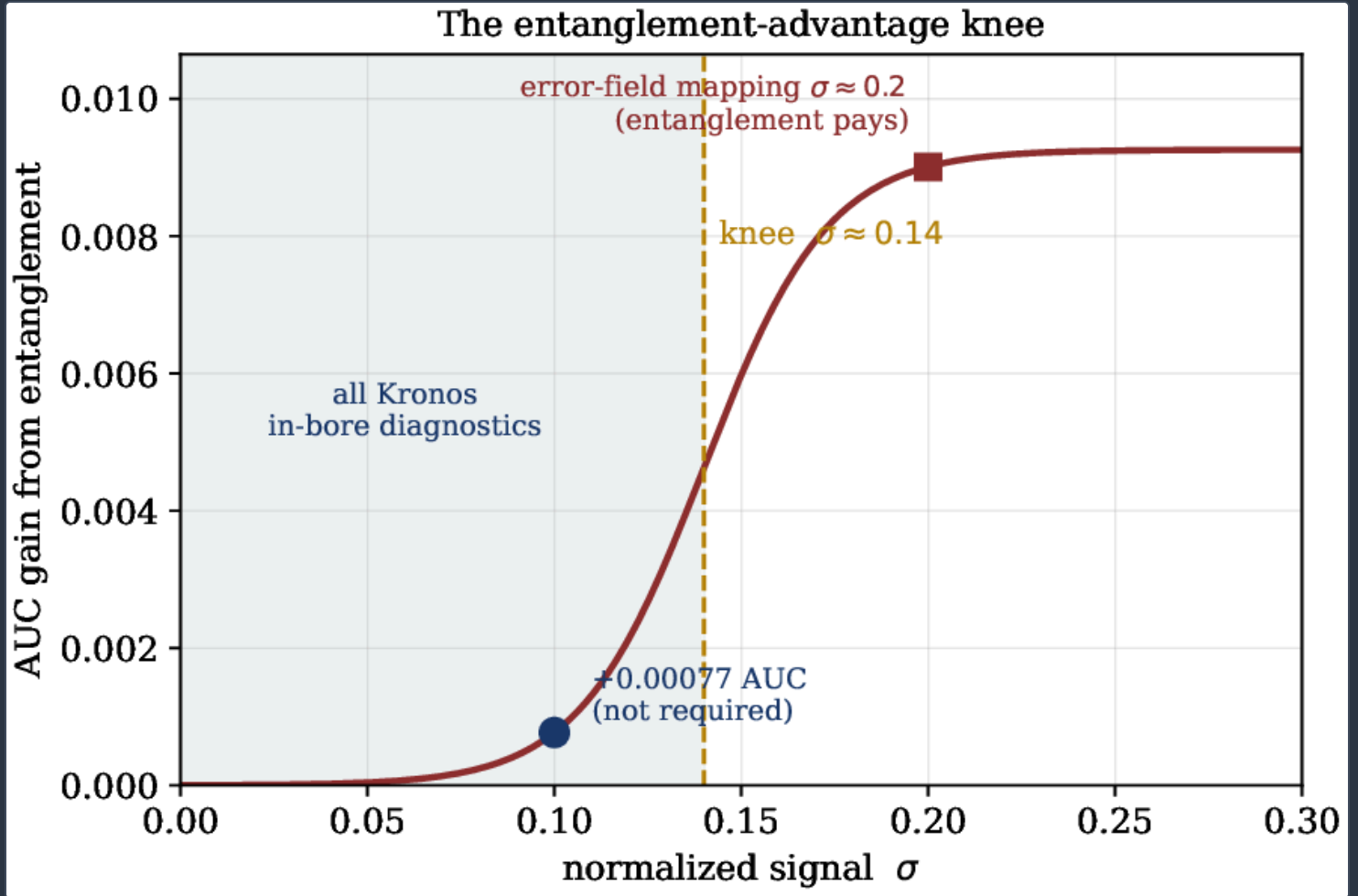
| SENSOR CLASS             | CLASS-TYPICAL SENSITIVITY | ROLE AT THE CENTREPOST                                    |
|--------------------------|---------------------------|-----------------------------------------------------------|
| NMR/EPR probe            | 0.01–1 ppm absolute       | primary absolute field monitor ( $\geq 100\times$ margin) |
| cryogenic Hall probe     | $\sim 100$ ppm–0.1 %      | engineering redundancy                                    |
| SQUID + pickup coil      | $\sim 1$                  | AC quench-precursor channel at standoff                   |
| NV-diamond ensemble      | $\sim 1$                  | quench / stray-field mapping; rad-hard, cryo-compatible   |
| atomic OPM (SERF/scalar) | 0.5–20                    | external error-field mapping only                         |

Sensor-class assignment for centrepost field monitoring and quench precursors (serial KX-L1-A18; class sensitivities from the literature <sup>[9,10,16,15]</sup>).

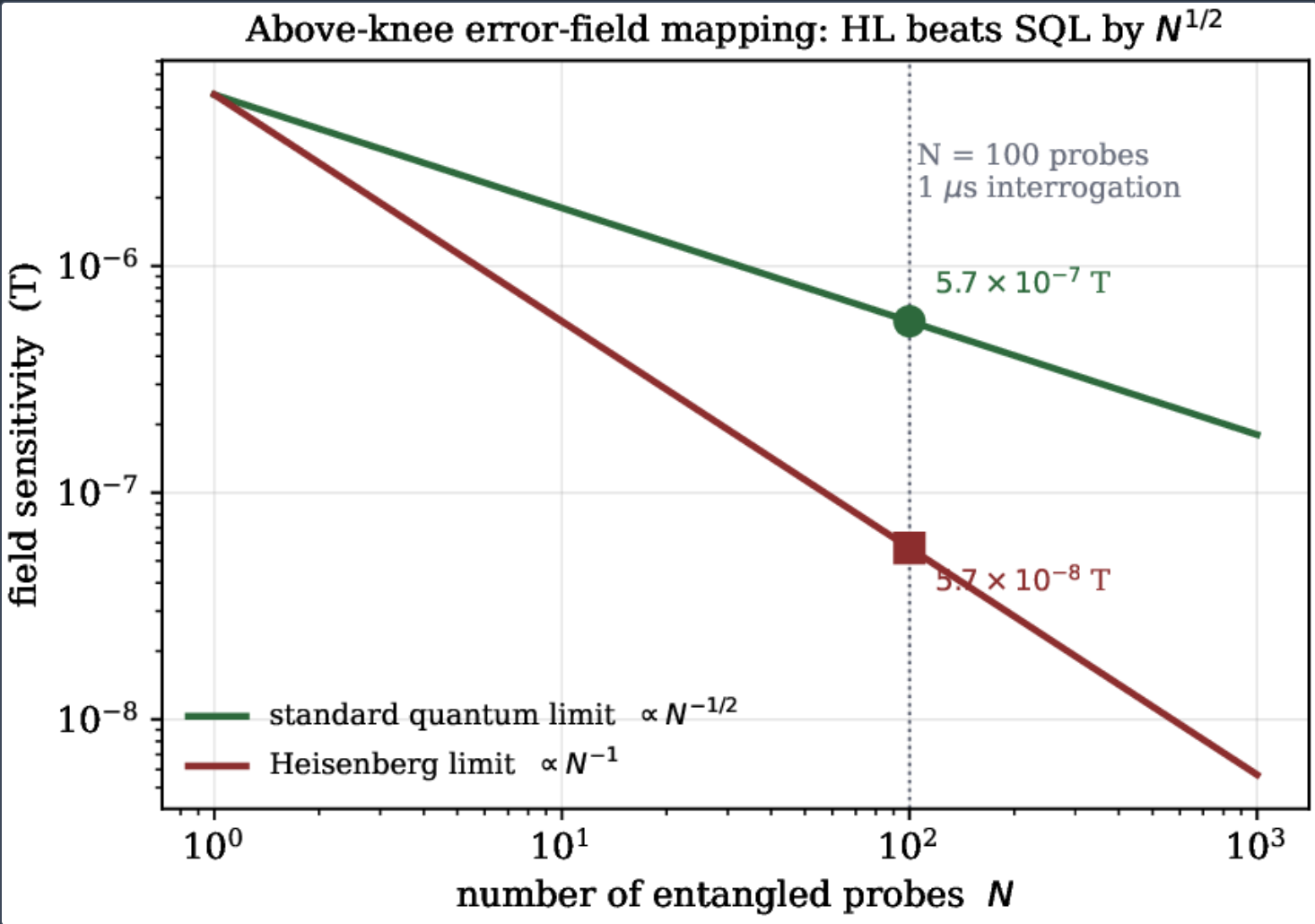


Sensor-class field resolution against the two centrepost tasks (serial KX-L1-A18): the absolute-monitor task (**1.68 mT**, dashed) and the quench precursor ( $\sim 1$  mT,  $5.9 \times 10^{-5}$  of the **16.84 T** peak, dotted). Every task is comfortably above the class floors; the NMR/EPR probe carries  $\geq 100\times$  margin on absolute monitoring.

The decisive quantity is where entanglement-enhanced protocols start to pay, and equation (11) answers it: the knee is at a normalised signal  $\sigma^* \approx 0.14$  (figure 5). The in-bore precursor and disruption-mode signals sit *below* the knee, so a coil-plus-SQUID or conventional NV readout already suffices and entanglement adds only  $+0.00077$  AUC — not worth its overhead. Above the knee, in deep sub-threshold error-field mapping ( $\sigma \approx 0.2$ ), the task is genuinely sensitivity-limited and entanglement helps: by equations (8)–(9) an NV ensemble running a Heisenberg-limited protocol carries the SQL sensitivity from  $5.7 \times 10^{-7}$  T down to  $5.7 \times 10^{-8}$  T at 100 probes and  $1 \mu\text{s}$  interrogation (figure 6, table 3). That is the one place quantum sensing earns its complexity, and the chain is specified to use it exactly there and nowhere else. The picotesla NV sensitivity of equation (10) is, notably, far finer than the millitesla precursor requires; its value is the margin it buys for multiplexing and for tolerating in-service degradation, the theme of the companion quench-detection paper <sup>[39]</sup>.



The entanglement-advantage knee (serial KX-L3-A5). The AUC gain attributable to entanglement is a threshold function of the normalised signal  $\sigma$  with the crossing at  $\sigma^* \approx 0.14$  (equation 11). All Kronos in-bore diagnostics sit in the shaded below-knee region, where the gain is a negligible  $+0.00077$  AUC; only above-knee error-field mapping ( $\sigma \approx 0.2$ ) rewards entanglement. Model curve pinned to the two frozen anchors.



Above-knee error-field mapping (serial KX-L3-A5). The standard-quantum-limit sensitivity (equation 7,  $\propto N^{-1/2}$ ) and the Heisenberg-limit sensitivity (equation 8,  $\propto N^{-1}$ ) are pinned at  $N = 100$  probes:  $5.7 \times 10^{-7}\ \text{T}$  and  $5.7 \times 10^{-8}\ \text{T}$  respectively, a factor  $\sqrt{N} = 10$  (equation 9).

| QUANTITY                             | VALUE                                               | BASIS               |
|--------------------------------------|-----------------------------------------------------|---------------------|
| centrepost peak field                | 16.84 T                                             | design point        |
| absolute-monitor task                | 1.68 mT                                             | $10^{-4}$ relative  |
| quench precursor                     | $\sim 1\ \text{mT}$ ( $5.9 \times 10^{-5}$ of peak) | winding signal      |
| entanglement knee                    | $\sigma^* \approx 0.14$                             | eq. (11)            |
| below-knee AUC gain                  | +0.00077 (negligible)                               | in-bore diagnostics |
| SQL sensitivity ( $N = 100$ )        | $5.7 \times 10^{-7}\ \text{T}$                      | eq. (7)             |
| Heisenberg sensitivity ( $N = 100$ ) | $5.7 \times 10^{-8}\ \text{T}$                      | eq. (8)             |

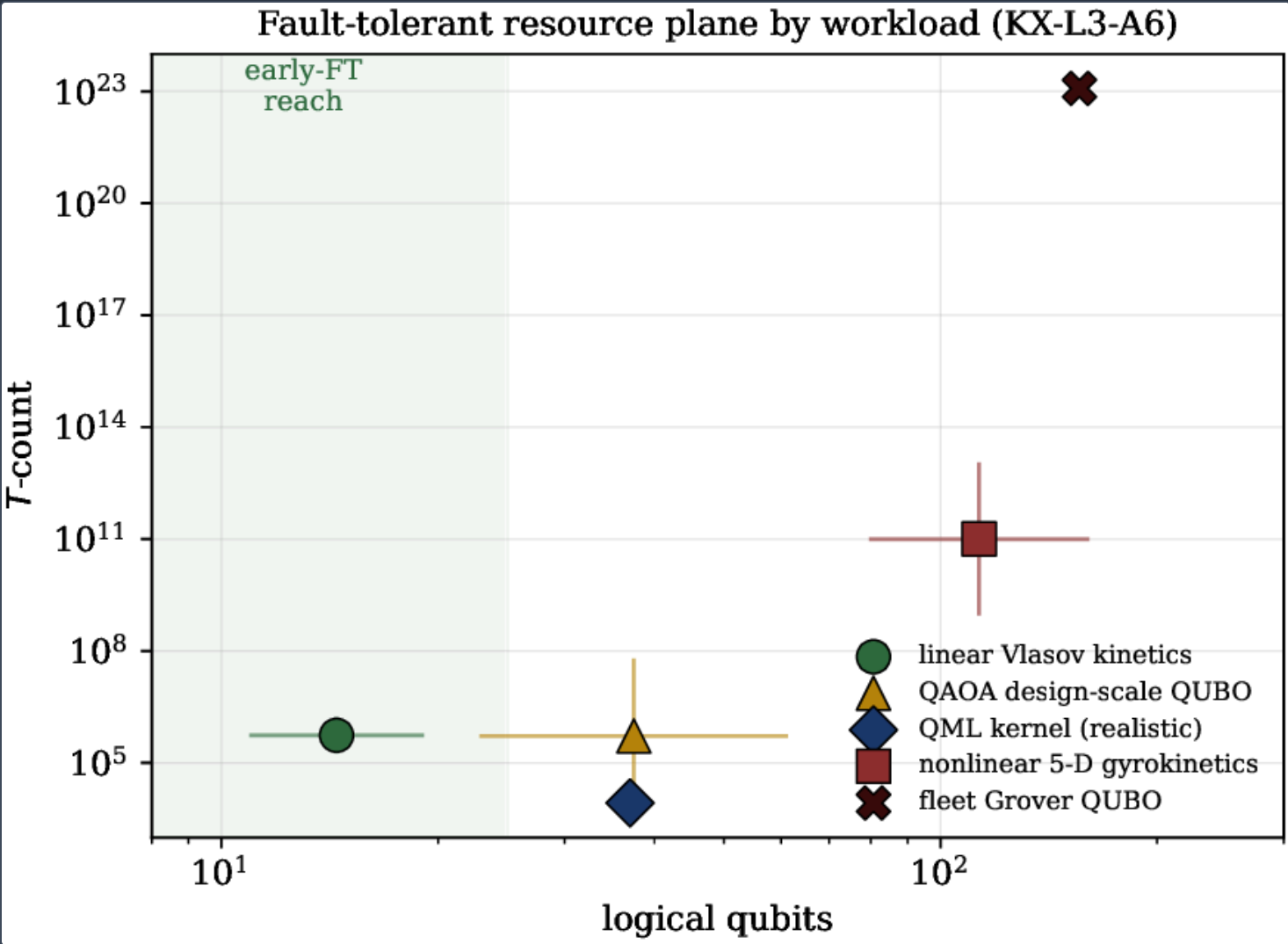
Entanglement-advantage accounting for centrepost sensing (serials KX-L1-A18, KX-L3-A5). The knee separates the below-knee regime, where every in-bore Kronos diagnostic sits, from the above-knee regime where a Heisenberg-limited protocol pays.

# Fault-tolerant resource estimates and the dated roadmap

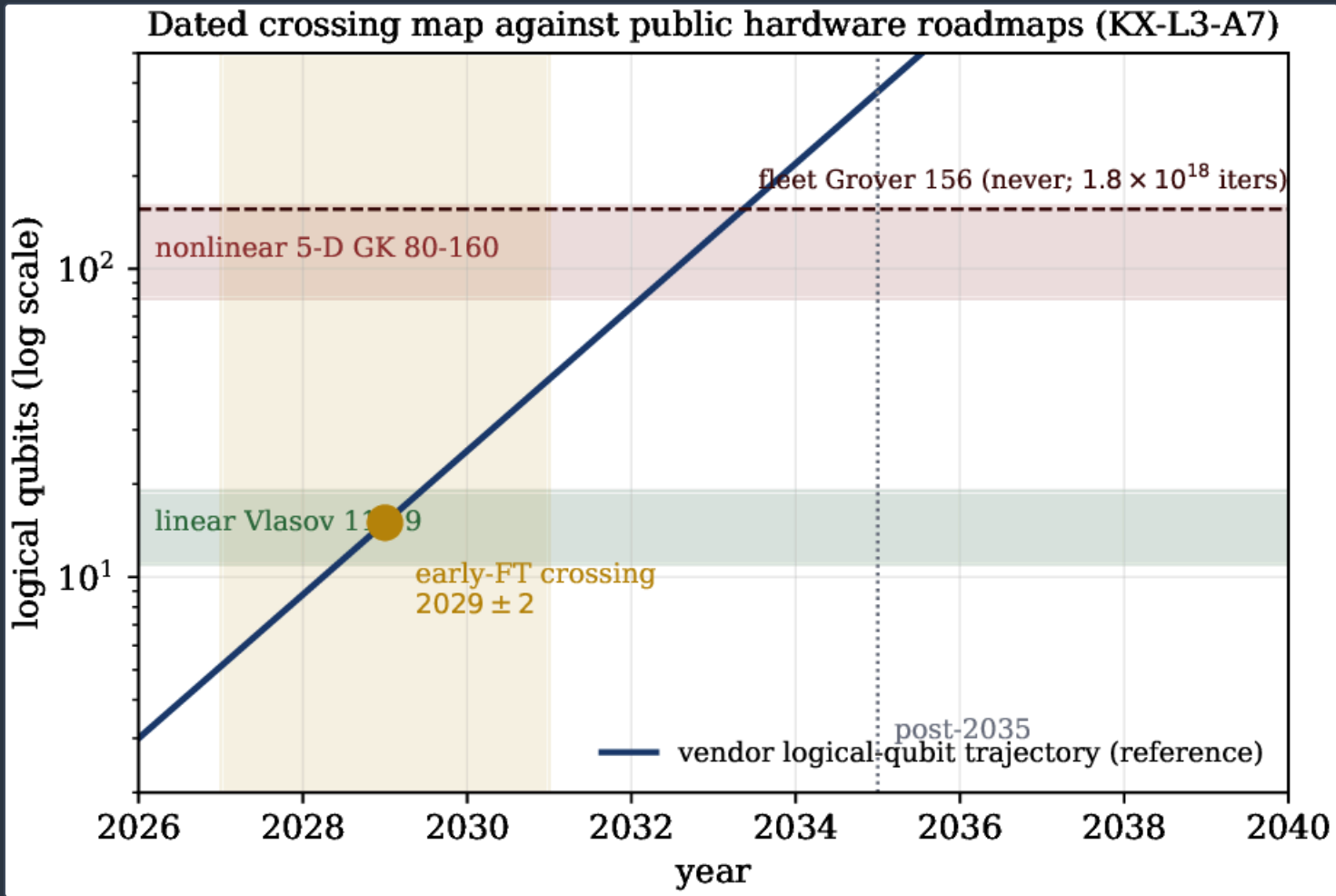
The final front is the eventual quantum solution of fusion-relevant equations. We recomputed the fault-tolerant resource ledger from the first formulas of §2 — the Grover iteration count of equation (12) and the surface-code footprint of equation (13) — across **50** rows and reproduced every value with zero discrepancy (serial KX-L3-A6). Table 4 states the thresholds by workload, figure 7 places them in the resource plane, and figure 8 dates them against public hardware roadmaps.

| WORKLOAD                   | LOGICAL QUBITS | $T$ -COUNT                          | CLASS / WINDOW                      |
|----------------------------|----------------|-------------------------------------|-------------------------------------|
| linear Vlasov kinetics     | 11–19          | $2 \times 10^5$ – $1.5 \times 10^6$ | early-FT, $\sim 2029 \pm 2$         |
| QAOA design-scale QUBO     | 23–61          | $5 \times 10^3$ – $5.5 \times 10^7$ | early-FT (classically trivial)      |
| QML kernel (realistic)     | 37             | $8.5 \times 10^3$ /circuit          | FT, no advantage                    |
| nonlinear 5-D gyrokinetics | 80–160         | $10^9$ – $10^{13}$                  | FT-horizon, post-2035               |
| fleet Grover QUBO          | 156            | $1.2 \times 10^{23}$                | never ( $1.8 \times 10^{18}$ iters) |

Fault-tolerant resource thresholds by workload and the earliest credible window (serial KX-L3-A6; logical-qubit and  $T$ -counts verified). The timeline (serial KX-L3-A7) is synthesized from public vendor roadmaps, which are references, not commitments.

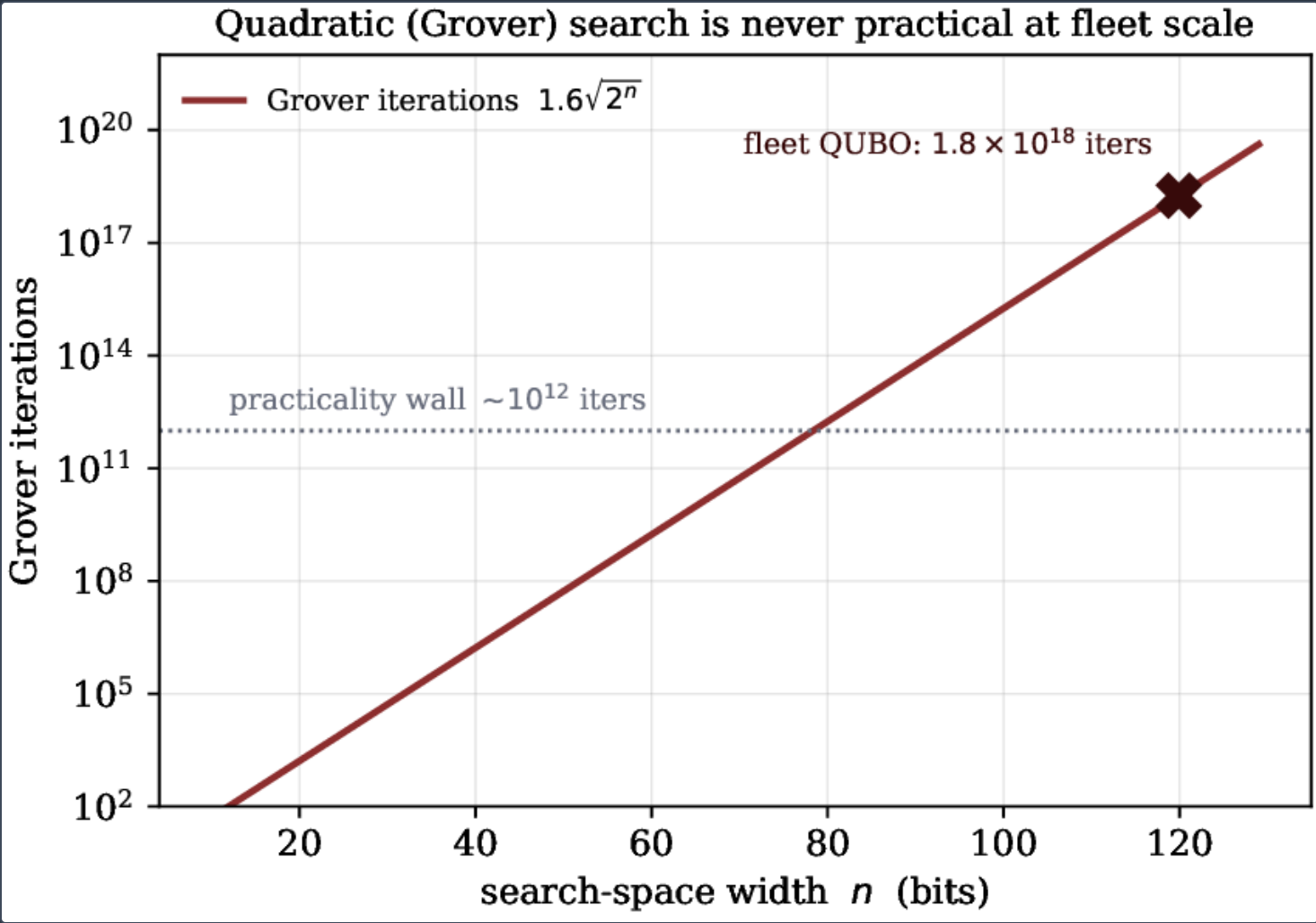


Fault-tolerant resource plane (serial KX-L3-A6): logical-qubit width against  $T$ -count for the five benchmarked workloads. Bars span the tabulated ranges. Only the linear-Vlasov and QAOA workloads fall within early-fault-tolerant reach (shaded); nonlinear gyrokinetics and fleet Grover sit many orders of magnitude beyond it.



Dated quantum-technology roadmap (serial KX-L3-A7). The vendor logical-qubit trajectory (reference) crosses the only near-term fusion-relevant threshold — linear Vlasov kinetics at 11–19 logical qubits — around  $2029 \pm 2$ . Nonlinear five-dimensional gyrokinetics (80–160 logical) lies beyond 2035, and fleet-scale Grover search ( $1.8 \times 10^{18}$  iterations) is never practical.

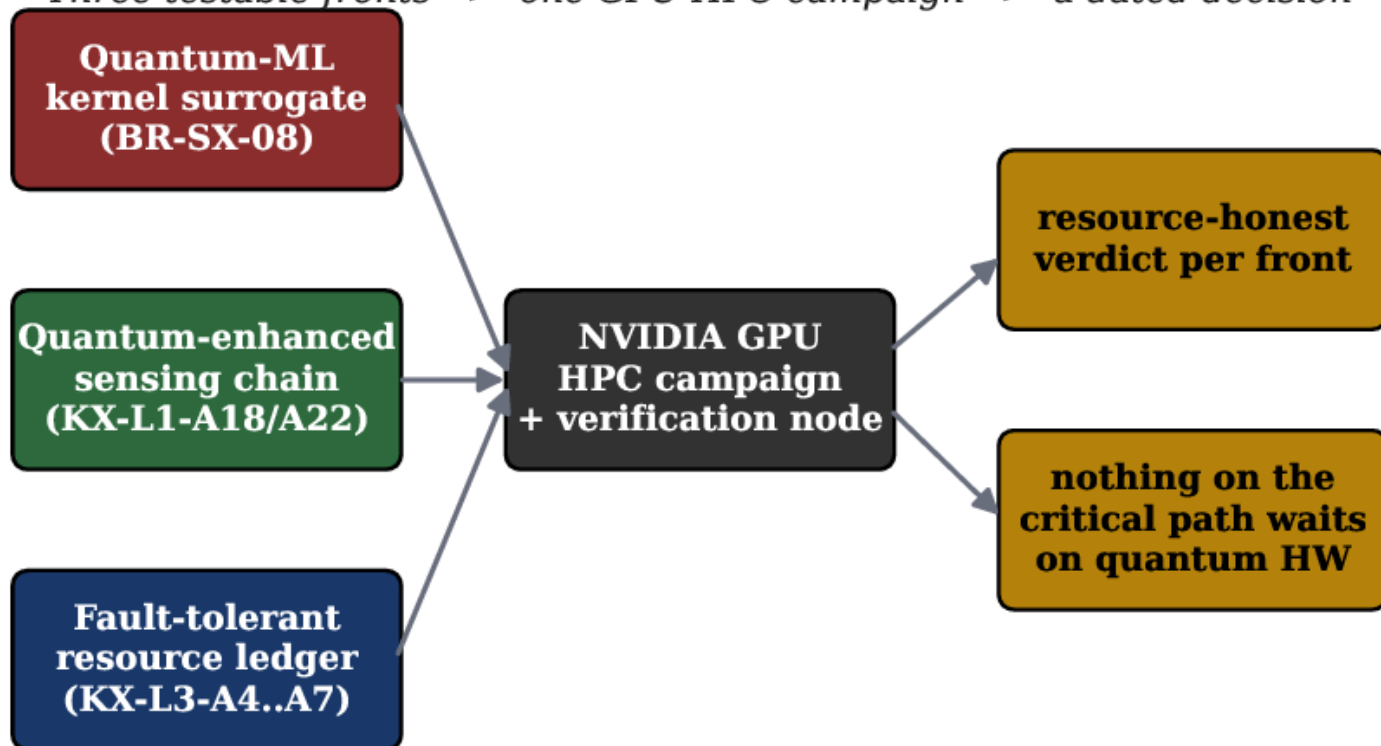
The map has exactly one dated, real opportunity: linear kinetic response on early-fault-tolerant machines around  $2029 \pm 2$ , an 11–19 logical-qubit workload that public roadmaps place within a credible window and that a Vlasov Hamiltonian-simulation encoding makes concrete <sup>[26]</sup>. Nonlinear five-dimensional gyrokinetic turbulence — the physics that actually sets confinement — remains a fault-tolerant-horizon problem beyond 2035 at  $10^9$ – $10^{13}$  *T*-gates, and fleet-scale Grover search stays prohibitive at any date: by equation (12) it demands  $1.8 \times 10^{18}$  iterations (figure 9), a textbook instance of Babbush *et al.*'s rule that quadratic speed-ups do not survive error-correction overheads <sup>[21]</sup>. Crucially, none of this is on the critical path: classical baselines of the CGYRO/GENE class <sup>[28,29]</sup> run the governing physics today, and a tensor-network kinetic solver offers a further classical route <sup>[40]</sup>, so the quantum opportunity is a validation-and-benchmark niche, not a design dependency.



Grover iteration count  $N_{\text{iter}} = 1.6\sqrt{2^n}$  (equation 12) against search-space width (serial KX-L3-A6). The fleet-scale QUBO sits at  $1.8 \times 10^{18}$  iterations, far above any practicality wall; a quadratic speed-up is worthless at this scale.

Figure 10 summarises the study: three testable fronts, one GPU-HPC campaign, and a dated decision on each.

*Three testable fronts -> one GPU-HPC campaign -> a dated decision*



(Schematic.) Study architecture. The quantum-ML surrogate, the quantum-sensing chain, and the fault-tolerant resource ledger are each computed on the NVIDIA GPU HPC campaign and a verification node, and each returns a resource-honest verdict; the common conclusion is that nothing on the design critical path waits on quantum hardware.

# Discussion

Across all three fronts the study delivers the same kind of value — a decision backed by numbers — and each number can be read against the published literature.

## QML IN CONTEXT

The measured AUC degradation ( $0.817 \rightarrow 0.691$ ,  $4 \rightarrow 8$  qubits) is not an implementation defect but a direct observation of the exponential kernel concentration proved by Thanasilp *et al.* <sup>[6]</sup> and consistent with the barren-plateau phenomenology of McClean *et al.* <sup>[7]</sup>. It is the data-side counterpart of Huang *et al.*'s finding that classical models learning from data are hard to beat on generic tasks <sup>[4]</sup>. The provable separations that do exist, such as Liu *et al.*'s discrete-log construction <sup>[5]</sup>, require adversarially structured data that transport classification does not supply; our negative result is therefore exactly the expected outcome for a natural-data problem, quantified on the Kronos features. This aligns with, and sharpens, the series' broader resource-honest computing assessment <sup>[34,35]</sup>: a quantum kernel is the wrong tool for this surrogate, and the neural-operator surrogate used elsewhere in the series is the right one.

## SENSING AGAINST THE QUANTUM-METROLOGY LITERATURE

The entanglement accounting rests on textbook bounds — the SQL and Heisenberg scalings of equations (7)–(8) <sup>[14,13]</sup> — and the sensor classes on mature device physics: ensemble NV magnetometry at  $\sim$  <sup>[10,11,12]</sup>, SQUIDS at the femtotesla floor <sup>[16]</sup>, and optical magnetometers likewise <sup>[15]</sup>. Our contribution is not a new device but a decision rule: the knee at  $\sigma^* \approx 0.14$  (equation 11) partitions the Kronos diagnostic set, placing every in-bore channel below the knee where entanglement is superfluous ( $+0.00077$  AUC) and reserving Heisenberg-limited protocols for the above-knee error-field-mapping task, where the  $\sqrt{N}$  gain of equation (9) is genuine. The burner counterpart of this analysis, for the **26.49 T** plug field, is carried in the companion burner quantum-stack paper <sup>[38]</sup>, and the NV-diamond quench-precursor lineage in the fused-detection paper <sup>[39]</sup>.

## RESOURCE ESTIMATES AGAINST THE FTQC LITERATURE

The ledger recomputation reproduces the surface-code cost model of Fowler *et al.* <sup>[19]</sup> and sits in the same estimation tradition as Gidney and Eker <sup>[22]</sup> and Beverland *et al.* <sup>[23]</sup>. Its dated conclusions are conservative and specific: the early-fault-tolerant linear-Vlasov niche at **11–19** logical qubits around **2029  $\pm$  2** matches the resource scale of the Engel–Smith–Parker Vlasov algorithm <sup>[26]</sup> and the broader plasma-quantum-computing survey of Dodin and Startsev <sup>[27]</sup>; the nonlinear-gyrokinetics horizon beyond **2035** reflects the  **$10^9$ – $10^{13}$   $T$** -gate cost; and the fleet-Grover verdict of “never” is the direct application of the quadratic-speed-up rule of Babbush *et al.* <sup>[21]</sup> through equation (12). The QAOA/QUBO design row, classically trivial at these sizes, is analysed in full in the companion quantum-optimisation paper <sup>[36]</sup>, and the linear-kinetics control-loop algorithms in <sup>[37]</sup>. The through-line is that quantum computing enters the Kronos program as a benchmark and validation resource on a dated horizon, never as a dependency of the frozen design point.

# Limitations

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The single open item is temporal rather than technical: the near-term quantum-computing niche — linear kinetic response on early-fault-tolerant hardware around **2029  $\pm$  2** — depends on vendor logical-qubit milestones that are referenced from public roadmaps and carry historical slippage risk. The date is therefore a reference window, not a Kronos commitment, and is re-checked at each annual review; because no design decision depends on it, slippage changes only the benchmark schedule, not the machine.

# Conclusion

We map, quantitatively, where quantum computing and sensing help a compact fusion program and where they do not. The quantum-kernel surrogate degrades with scale (AUC  $0.817 \rightarrow 0.691$ ,  $4 \rightarrow 8$  qubits) through the exponential kernel concentration of equation (4) and is  $9\text{--}11$  orders slower than classical at fault-tolerant scale; the sensing chain for the  $16.84\text{ T}$  centrepost uses NMR/EPR, SQUID and NV-diamond channels, with entanglement reserved for above-knee error-field mapping ( $\sigma^* \approx 0.14, 5.7 \times 10^{-8}\text{ T}$  at  $100$  probes) and never conflated with the distinct  $26.49\text{ T}$  burner plug; and the verified  $50$ -row fault-tolerant ledger dates one near-term niche — linear kinetics around  $2029 \pm 2$  — with nonlinear gyrokinetics beyond  $2035$  and fleet Grover never. Nothing on the design critical path waits on quantum hardware, and establishing that precisely, with the mathematics and the numbers behind it, is the result.

**Acknowledgments** Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

**Reproducibility.** The quantum-ML scaling result, the sensing specification, and the fault-tolerant thresholds are frozen entries (serials BR-SX-08, KX-L1-A18, KX-L1-A22, KX-L3-A4, KX-L3-A5, KX-L3-A6, KX-L3-A7) in the Kronos Physics De-Risking Register <sup>[33]</sup>; the  $50$ -row resource recomputation reproduced every value from the stated formulas with zero failures, and the statevector kernel benchmark and analytic sensing model are regenerable from the figure-generation script deposited with this paper. This paper is archived at its DOI [doi:10.5281/zenodo.22645905](https://doi.org/10.5281/zenodo.22645905) and at its official page <https://www.kronosfusionenergy.com/publications/quantum-machine-learning-surrogates-entanglement-enhanced.html>; the register is archived at [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057).

**Data availability** The quantum-kernel benchmark, sensor-requirement tables, entanglement-knee sweep, and resource-recomputation ledger used for the figures are archived with the deposit at [doi:10.5281/zenodo.22645905](https://doi.org/10.5281/zenodo.22645905), which references the Kronos 2026 series including the AI-and-quantum-computing paper <sup>[34]</sup>. All quantitative values trace to the register <sup>[33]</sup> by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

**Competing interests** Provisional patent applications relating to aspects of this work have been filed.

**References**

A P P A R A T U S

## *Portfolio, People & Record*

*The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

|               |                                                                                   |         |
|---------------|-----------------------------------------------------------------------------------|---------|
| US 12,009,112 | High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550 | GRANTED |
| US 17/878,507 | Core device / method (KRONO-001A) · utility                                       | PENDING |

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

|            |                                                                                                                                                                                                                                |       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 64/124,209 | Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026                                                                                                                                             | FILED |
| 64/124,220 | Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026                                                                                                                             | FILED |
| 64/115,167 | KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026                                                                                                                                                          | FILED |
| 64/005,440 | Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026                                                                                                                       | FILED |
| 64/105,530 | Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026                                                                                                                                | FILED |
| 64/128,097 | Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop | FILED |

TRADEMARK & ORIGIN

|              |                                                                          |          |
|--------------|--------------------------------------------------------------------------|----------|
| FTK-98801894 | Kronos Fusion Energy — trademark application                             | FILED    |
| KR-2022-★    | Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022 | PRIORITY |

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any  $Q > 1$  milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

## THE TEAM

# Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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